THOMAS MASON

Applied Economist • Senior Lecturer (Associate Professor, with tenure) in Health Economics, Lancaster University

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I am an applied economist, and my research examines the effects of policy interventions on health and economic outcomes: how policies and organisational arrangements change the behaviour and outcomes of individuals and institutions, estimated using microeconometric methods applied to large administrative and survey data. My research spans the health care labour market, the design of provider incentives, regional economic policy, and inequalities, in a sector that accounts for around one in eight UK jobs and is a primary setting for how AI reshapes work. It has been carried directly into government decision-making, including resource allocation formulae adopted for England's public health grant from 2016.

Research

- **Health care labour market:** the demand for and supply of labour, workforce planning, the substitution of tasks between types of labour (skill mix), and the implications of population ageing and morbidity for the need for care.
- **Incentives and provider behaviour:** pay-for-performance as principal–agent contracts, and their unintended consequences across public services.
- **Regional policy and inequalities:** regional economic policy evaluation (synthetic control), needs-based resource allocation, and inequalities in chronic disease.
- **Microeconometric methods:** causal inference (difference-in-differences, event studies, synthetic control, panel data, instrumental variables, matching) on large administrative data (HES, CPRD, NDTMS) and linked survey sources; evidence synthesis.
- **Software:** Stata (advanced, 15 years; large-scale data pipelines); Python-based analysis via AI-assisted workflows (Claude Code) in daily use; SQL (developing).

Selected publications

1. * Exploring the limitations of age-based models for health care planning. *Social Science & Medicine*, 2015.
2. * Did paying drugs misuse treatment providers for outcomes lead to unintended consequences for hospital admissions? Difference-in-differences analysis of a pay-for-performance scheme in England. *Addiction*, 2021.
3. Measuring performance in cancer services during COVID-19 and beyond: a novel case throughput metric using NHS Cancer Waiting Times data. *Journal of Public Health*, 2026.
4. Will the need-based planning of health human resources currently undertaken in several countries lead to excess supply and inefficiency? A comment on Basu and Pak. *Health Economics*, 2017.
5. Investigating the relationship between formal and informal care: an application using panel data for people living together. *Health Economics*, 2019.
6. * The impact of paying treatment providers for outcomes: difference-in-differences analysis of the 'payment by results for drugs recovery' pilot. *Addiction*, 2015. *(Shortlisted, EMCDDA Scientific Committee award, 2016.)*
7. * Is the distribution of care quality provided under pay-for-performance equitable? Evidence from the Advancing Quality programme in England. *International Journal for Equity in Health*, 2016.
8. Development of a resource allocation formula for substance misuse treatment services. *Journal of Public Health*, 2018.
9. Using machine learning to predict anticoagulation control in atrial fibrillation: a UK Clinical Practice Research Datalink study. *Informatics in Medicine Unlocked*, 2021.

* denotes first-authored. Full list of 30 publications at `orcid.org/0000-0003-3135-0364`.

Employment

Senior Lecturer (Associate Professor, with tenure) in Health Economics, Lancaster University 2022–present

- Lead of the quantitative (synthetic control) work package on a £1.6m **NIHR evaluation of community wealth building**, a regional economic policy, comparing Scottish and English localities.
- Co-lead, drug and alcohol theme, **NIHR Mental Health Research Group** (£11.3m); co-investigator on NIHR studies of primary care nurse continuity (£100k) and of quantitative evidence in public health commissioning (£50k).
- **Co-Director of Research**, Division of Health Research (a research-intensive department of around 150 staff).
- Module lead, Applied Health Economics; supervise part-time PhD students (completed doctoral supervision at Manchester).

Research Fellow (Health Economics) and Lecturer (Economics), University of Manchester 2020–2022

- Lectured on the MSc Economics programme (Economics of Health). Evaluations of the NHS Diabetes Prevention Programme and of a hospital domestic violence advisor service that informed its permanent commissioning.

Senior Research Analyst, HEOR Ltd (private-sector consultancy) 2018–2020

- Led a team of three health economists delivering commissioned research; lead economist on a Bristol-Myers Squibb study developing a machine-learning algorithm to predict anticoagulation control in patients with atrial fibrillation, using linked CPRD–HES data.

Research Associate, then Research Fellow, University of Manchester 2010–2018

- Difference-in-differences evaluations of pay-for-performance as principal–agent contracts; conceptual and forecasting models of health care labour demand and supply. Graduate Teaching Assistant, Econometrics (BSc), 2011–2015: departmental teaching award (top 5% of tutors).

Education

BA(Hons) Economics and Politics, MSc Economics, PhD (University of Manchester).
PhD thesis: "Substance misuse related hospital admissions, costs and treatment outcomes: econometric analysis of administrative data for England".
Recent professional development: CodeChella workshop on difference-in-differences and the use of Claude Code in economic research (Mixtape Sessions, Madrid, 2026).

Research funding

Co-investigator on four current NIHR awards totalling over £13m (2025–2026): NIHR Mental Health Research Group, £11.3m (co-lead, drug and alcohol theme); an evaluation of regional economic policy (community wealth building), £1.6m (synthetic control work-package lead); NIHR Three Schools, £100k (effects of primary care nurse continuity on patient outcomes, using linked CPRD–HES data); NIHR Research Support Service Methodology Fund, £50k. Previously co-investigator or lead on funded research totalling over £1.2m (Medical Research Council, Department of Health, NHS England, Bristol-Myers Squibb).

Leadership, policy impact and external roles

- **Previous leadership:** Deputy theme lead for Health, Lancaster Data Science and AI Institute; health-theme lead, ESRC North West Social Science Doctoral Training Partnership.
- **Policy impact:** the public health resource allocation formulae I co-developed were adopted by the Department of Health as the basis for target allocations of the public health grant from 2016; research commissioned by and presented to NHS England, the Department of Health and the Home Office.
- **Affiliations and external roles:** member, NIHR Public Health Research Programme Funding Committee; Affiliate, The Work Foundation (Lancaster University), a labour-market and employment think tank; methods expert, NIHR Research Support Service; expert peer reviewer (Home Office; Oak Foundation).